

Questions - Part 11 - Science II

A Quick-Reference Guide

Science Questions 2 - Statistical Significance & Other Data Contexts

Interpretation is important and not always clear or easy.

In scientific writing, and in non-technical writing, there can be a lack of clarity in correcting understanding the situation being described for assorted reasons: sloppy word choice; deception; lack of detail; unclear references and associations; blending of categories, etc.

Contexts including statistical significance

1) Clearly good or clearly bad 2) Strong, weak, no statistical correlation 3) Positive or negative/inverse correlation 4) Large or small effect size 5) Big or small data set	6) High or low relevance or importance (for a given context) 7) Target market and profitability 8) Dramatic, advertising, PR effect 9) Development practicality - time-table, likelihood of moving through R&D
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Forms of questions and answers regarding statistical significance

The baseline standard question being: How statistically significant is your correlation?

Q: What kind of correlation is it? A: There is a (positive/negative) correlation.	Q: What does this graph show? A: There is a positive/negative correlation between X & Y.	Q: What is the relationship between X and Y? A: X correlates positively/negatively with Y.
Q: What is the relationship here? A: X is proportional to Y.	Q: Does X correspond to Y? A: X corresponds to/with Y.	Q: Is X correlated with Y? A: X is correlated with Y.
Q: Is X proportional to Y? Q: What is proportional to what? A: X is inversely proportional to Y.	Q: Is X inversely correlated with Y? A: X is inversely correlated with Y. A: X has an inverse correlation with Y.	Q: Does X correlate positively with Y? A: X correlates positively with Y. A: X correlates negatives with Y.
Q: How statistically significant is the correlation? A: It is not significant. A: It is nearly significant. A: It is significant. A: It is highly significant.	Q: There is a negative/positive correlation between x and y? A: There is a negative/positive correlation between x and y. A: Here we have a negative/positive correlation between x and y. A: This shows a negative/positive correlation between x and y.	For no correlation: A: X is not correlated with Y. A: X is not dependent on Y. A: X is not related to Y. A: X is not related to Y.

On "Proof":

Any mention of words such as proof or certainty have no direct connection to the topic of statistically significant correlations, unless of course the words are used in a caveat about their absence. Such words may indicate whether the text in question is clearly written.